MOUNTAINBOARD • T4 STUDY EDITION

固定器锚盘
Binding Anchor

原教材第24–54页 · 中英对照版
Workbook pages 24–54 · Bilingual edition

31页原文完整保留，每页与中文译文并排对照。
All 31 original pages are retained and paired with Chinese translation.

阅读说明 / Reading notes

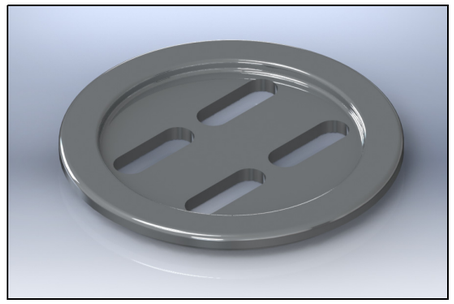
- 本范围是用户指定的T4作业；原教材的章节标题为“Lesson 2: Basic Functionality / 第2课：基本功能”。
- 左侧保留原英文页面的全部文字、工程图、操作截图和图片；右侧为中文翻译，命令名称尽量中英并列。PDF左侧保留原始矢量页面；Word左侧为180 dpi页面图像，右侧中文可编辑。
- 数值及教材内部差异按原文保留，并明确标注译注。第28页概述的Ø75与第29页明确输入的Ø82不一致；第24页最终工程图外径为Ø82。第51页正文2015 Alloy与第52页截图2014 Alloy也不一致。
- 此为学习译本；译注与补充图示标签采用明确标记，原始页面未被修改。原文中的文件路径、旧模板与界面截图属于教材原版内容；SOLIDWORKS 2026操作教学见配套网站。

Source: Mountainboard_Workbook_Pages_24-54.pdf
Layout: English original on the left / Chinese translation on the right

ENGLISH ORIGINAL

Active Learning Exercise, Part 1 — Creating a Basic Part

The first part created will be the `Binding Anchor` shown at right. We will use SolidWorks to create this part.

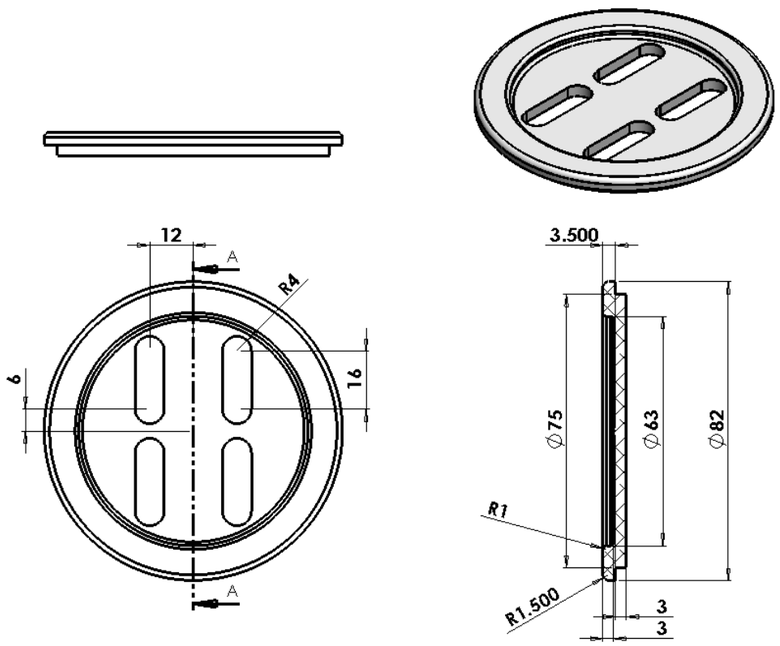


Design Intent

Before starting on the actual steps to create the `Binding Anchor`, we need to determine the design intent. This is a list of requirements the finished part needs to meet. The design intent will tell us what the finished part must be able to do.

- ❑ The `Binding Anchor` will position the binding on the deck.
- ❑ The `Binding Anchor` must allow the binding to be positioned both along the centerline of the deck as well as adjusting the angle to the deck to allow the rider to set a comfortable stance.
- ❑ The `Binding Anchor` clamps the binding to the deck.
- ❑ There must be no sharp edges to injure a rider.

The `Binding Anchor` will look like the drawing below. Step-by-step instructions are given below.



中文译文 / CHINESE TRANSLATION

主动学习练习，第 1 部分——创建基本零件/Active Learning Exercise, Part 1 — Creating a Basic Part

我们首先要创建的零件是右图所示的固定器锚盘/Binding Anchor。我们将使用 SOLIDWORKS 创建这个零件。

设计意图/Design Intent

在开始执行创建固定器锚盘/Binding Anchor 的具体步骤之前，需要先确定设计意图。设计意图是一组最终零件必须满足的要求，它告诉我们完成后的零件必须具备哪些功能。

• 固定器锚盘/Binding Anchor 将固定器定位在板面/Deck 上。

• 固定器锚盘/Binding Anchor 必须既能使固定器沿板面的中心线调整位置，也能调整固定器与板面之间的角度，让骑行者设定舒适的站姿。

• 固定器锚盘/Binding Anchor 将固定器夹紧在板面上。

• 不得有可能伤及骑行者的锐利边缘。

固定器锚盘/Binding Anchor 的外形将如下方工程图所示。后续内容提供逐步操作说明。

工程图尺寸标注：外径 Ø82；相关圆直径 Ø75 和 Ø63；标注尺寸 12、16、6、3.500、3、3；半径 R4、R1、R1.500；剖切标识 A—A。所有数值与原图一致，具体尺寸引线、剖视及位置关系以保留的原图为准。

ENGLISH ORIGINAL

Task 1— Create a New Part Document

1 Create a new part. Click **New** on the Standard toolbar.

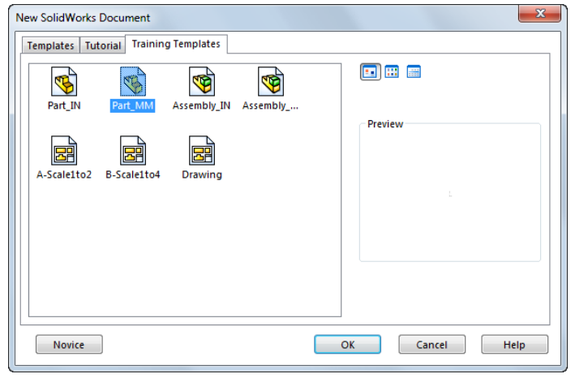
The **New SolidWorks Document** dialog box appears.

2 Click the **Training Templates** tab.

3 Select the **Part_MM** icon.

4 Click **OK**.

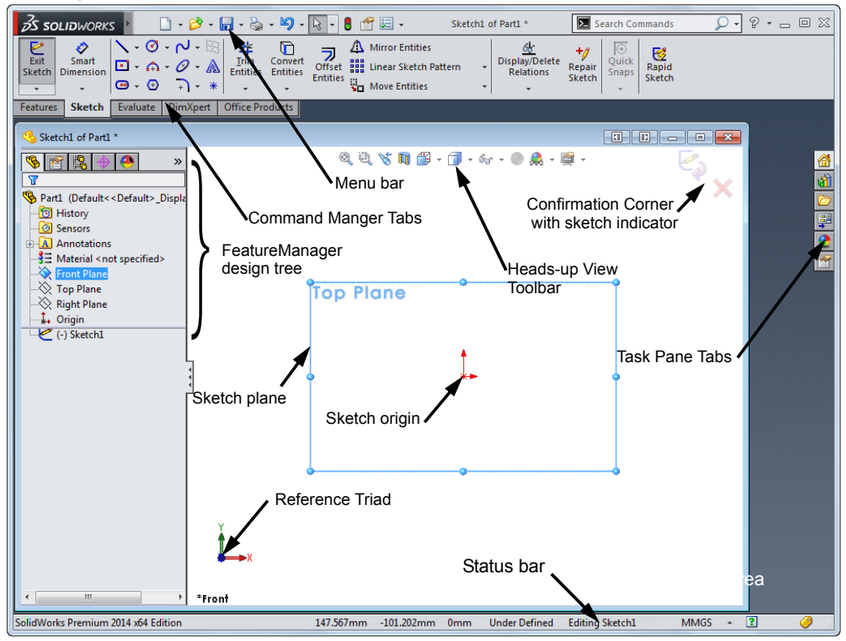
A new part document window appears.



Overview of the SolidWorks Window

When you create a new sketch:

- A sketch origin appears in the center of the graphics area.
- "Editing Sketch" appears in the status bar at the bottom of the screen.
- `Sketch1` appears in the FeatureManager design tree.
- The status bar shows the position of the pointer, or sketch tool, in relation to the sketch origin.



中文译文 / CHINESE TRANSLATION

任务 1——创建新的零件文档/Task 1 — Create a New Part Document

1 创建一个新零件。单击标准工具栏/Standard toolbar 上的新建/New。出现新建 SOLIDWORKS 文档/New SOLIDWORKS Document 对话框。

2 单击培训模板/Training Templates 选项卡。

3 选择 Part_MM 图标。

4 单击确定/OK。出现一个新的零件文档窗口。

SOLIDWORKS 窗口概览/Overview of the SOLIDWORKS Window

创建新草图时：

• 图形区域中央出现一个草图原点/Sketch origin。

• 屏幕底部的状态栏/Status bar 中显示正在编辑草图/Editing Sketch。

• FeatureManager 设计树/FeatureManager design tree 中出现草图1/Sketch1。

• 状态栏显示指针或草图工具相对于草图原点的位置。

窗口示意图标注：菜单栏/Menu bar；FeatureManager 设计树/FeatureManager design tree；草图基准面/Sketch plane；状态栏/Status bar；图形区域/Graphics area；草图原点/Sketch origin；前导视图工具栏/Heads-up View Toolbar；带草图指示符的确认角/Confirmation Corner with sketch indicator；参考三重轴/Reference Triad；CommandManager 选项卡/Command Manager Tabs；任务窗格选项卡/Task Pane Tabs。

新建文档对话框中可见的其他标签：模板/Templates；教程/Tutorial；预览/Preview；初学者/Novice；取消/Cancel；帮助/Help。模板名 Part_IN、Part_MM、Assembly_IN、Drawing 等保留原文件名。

ENGLISH ORIGINAL

First Feature

The first feature requires:

- Sketch plane – Top
- Sketch profile – 2D Circle
- Feature type – Extruded boss feature

Sketching verses Drawing

The basis of most SolidWorks features is the sketch. Sketching is different from drawing in that drawings are created to the correct size as the lines and circles are drawn on the screen. With sketches, you only get the lines and circles close to their correct size. Dimensions and relationships will be added to make the sketch the correct size.

Open a Sketch

5 In the FeatureManager design tree, select (click once) the Top plane.

6 Select the **Sketch** tab on the Command Manager.

7 Open a 2D sketch. Click **Sketch** on the Command Manager.

The sketch opens on the Top plane.

中文译文 / CHINESE TRANSLATION

第一个特征/First Feature

第一个特征需要：草图基准面/Sketch plane——上视基准面/Top；草图轮廓/Sketch profile——二维圆/2D Circle；特征类型/Feature type——拉伸凸台特征/Extruded boss feature。

草图绘制与精确制图/Sketching versus Drawing

大多数 SOLIDWORKS 特征以草图为基础。绘制草图与精确制图有所不同：精确制图是在屏幕上绘制直线和圆的同时就使它们具有正确尺寸；绘制草图时，只需将直线和圆画成接近正确的大小，然后再添加尺寸和几何关系，使草图达到准确尺寸。

打开草图/Open a Sketch

5 在 FeatureManager 设计树/FeatureManager design tree 中选择上视基准面/Top plane，单击一次即可。

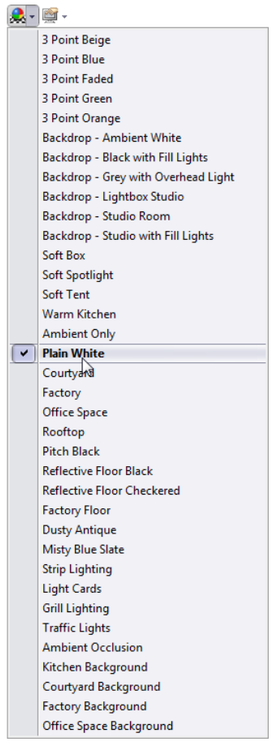
6 选择 CommandManager 上的草图/Sketch 选项卡。

7 打开一个二维草图：单击 CommandManager 上的草图/Sketch。草图将在上视基准面/Top plane 上打开。

ENGLISH ORIGINAL

Background

To make the images in this course easier to read, we will use a white background instead of the default. To change the background scene, click **Apply Scene** on the **Heads-up View** toolbar and select the scene you want to use.



View Orientation

When we open a sketch for the first feature, SolidWorks will automatically change the view orientation to be normal to the sketch plane. This makes it easier to see the sketch. It is like looking straight down on a piece of paper.

Confirmation Corner

When many SolidWorks commands are active, a symbol or a set of symbols appears in the upper right corner of the graphics area. This area is called the **Confirmation Corner**.

Sketch Indicator

When a sketch is active, or open, a symbol appears in the confirmation corner that looks like the **Sketch** tool. It provides a visual reminder that you are active in a sketch. Clicking the symbol exits the sketch saving your changes. Clicking the red X exits the sketch discarding your changes.

When other commands are active, the confirmation corner displays two symbols: a check mark and an X. The check mark executes the current command. The X cancels the command.

Sketch Entities

SolidWorks provides a variety of tools to create sketches. They can be found on both the **Sketch Entity** menu and most can also be found on the **Sketch** toolbar.

中文译文 / CHINESE TRANSLATION

背景/Background

为了使本课程中的图像更清晰易读，我们使用白色背景代替默认背景。若要更改背景场景，单击前导视图工具栏/Heads-up View toolbar 上的应用布景/Apply Scene，然后选择要使用的场景。

视图定向/View Orientation

打开用于创建第一个特征的草图时，SOLIDWORKS 会自动将视图方向切换为垂直于草图基准面。这有助于观察草图，就像正对着一张纸向下看。

确认角/Confirmation Corner

在许多 SOLIDWORKS 命令处于激活状态时，图形区域的右上角会显示一个或一组符号。这个区域称为确认角/Confirmation Corner。

草图指示符/Sketch Indicator

草图处于激活或打开状态时，确认角内会出现一个类似草图/Sketch 工具的符号。它以视觉方式提醒你当前正在编辑草图。单击该符号会退出草图并保存更改；单击红色 X 会退出草图并放弃更改。

其他命令处于激活状态时，确认角会显示两个符号：勾选标记和 X。勾选标记用于执行当前命令，X 用于取消命令。

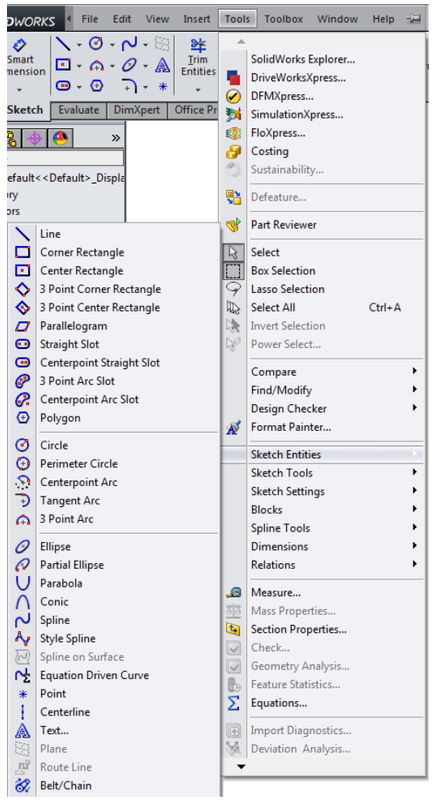
草图绘制实体/Sketch Entities

SOLIDWORKS 提供了多种创建草图的工具。可在草图绘制实体/Sketch Entities 菜单中找到这些工具，其中大多数也可在草图/Sketch 工具栏上找到。

ENGLISH ORIGINAL

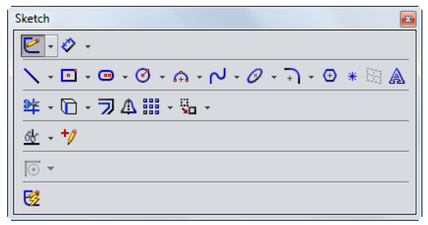
Sketch Menu

The Sketch Tools menu is found by clicking **Tools, Sketch Entities**. All of the sketch tools are listed in the menu.



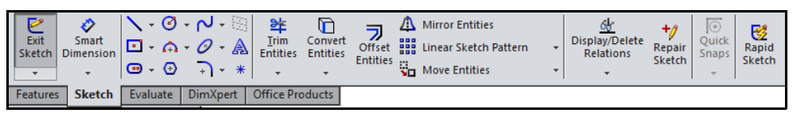
Sketch Toolbar

The Sketch Toolbar contains most of the sketch entities. It can be customized by adding or removing buttons.



Command Manager

Selecting the Sketch tab on the Command Manager will display the sketch tools.



Task 2 — Create the first sketch

The first feature will be a short cylinder, 75mm in diameter and 3.5mm thick.

The Circle

The circle tool creates 2D circles. Using the mouse, press the left mouse button at the location for the center of the circle, then (holding down the left mouse button) drag until the circle is approximately the correct size. Release the left mouse button.

中文译文 / CHINESE TRANSLATION

草图菜单/Sketch Menu

依次单击工具/Tools、草图绘制实体/Sketch Entities，即可找到草图工具菜单。所有草图工具均列在该菜单中。

草图工具栏/Sketch Toolbar

草图/Sketch 工具栏包含大多数草图绘制实体工具。可通过添加或删除按钮自定义此工具栏。

CommandManager

选择 CommandManager 上的草图/Sketch 选项卡，将显示草图工具。

任务 2——创建第一个草图/Task 2 — Create the first sketch

第一个特征是一个短圆柱，直径 75 mm、厚度 3.5 mm。

圆/The Circle

圆/Circle 工具用于创建二维圆。使用鼠标，在圆心位置按下鼠标左键，然后保持按住左键并拖动，直到圆的大小接近所需尺寸，最后松开鼠标左键。

【译注：本页原文写直径 75 mm；第 29 页明确要求输入直径 82 mm，第 24 页最终工程图的外径也为 Ø82。本译文保留原文差异，建模时应结合后续步骤及工程图核对。】

草图绘制实体菜单中的工具名称：直线/Line；边角矩形/Corner Rectangle；中心矩形/Center Rectangle；三点边角矩形/3 Point Corner Rectangle；三点中心矩形/3 Point Center Rectangle；平行四边形/Parallelogram；直槽口/Straight Slot；中心点直槽口/Centerpoint Straight Slot；三点圆弧槽口/3 Point Arc Slot；中心点圆弧槽口/Centerpoint Arc Slot；多边形/Polygon；圆/Circle；周边圆/Perimeter Circle；圆心圆弧/Centerpoint Arc；切线弧/Tangent Arc；三点圆弧/3 Point Arc；椭圆/Ellipse；部分椭圆/Partial Ellipse；抛物线/Parabola；圆锥曲线/Conic；样条曲线/Spline；样式曲线/Style Spline；曲面上的样条曲线/Spline on Surface；方程式驱动的曲线/Equation Driven Curve；点/Point；中心线/Centerline；文字/Text；基准面/Plane；步路线/Route Line；皮带/链/Belt/Chain。

CommandManager 截图中的文字工具：退出草图/Exit Sketch；智能尺寸/Smart Dimension；剪裁实体/Trim Entities；转换实体引用/Convert Entities；等距实体/Offset Entities；镜向实体/Mirror Entities；线性草图阵列/Linear Sketch Pattern；移动实体/Move Entities；显示/删除几何关系/Display/Delete Relations；修复草图/Repair Sketch；快速捕捉/Quick Snaps；快速草图/Rapid Sketch。

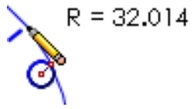
ENGLISH ORIGINAL

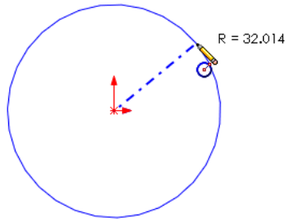
1 Click the **Circle** tool from the Sketch toolbar. The curser will show that the circle tool is active by displaying a circle under the drawing tool .

2 Move the cursor over the origin until an orange circle appears. The little yellow icon below the drawing tool will show that we are going to make the center of the circle coincident with the sketch origin.



3 Press the left mouse button and drag the circle until it is just about 32 mm. The cursor feedback will show the radius of the circle.



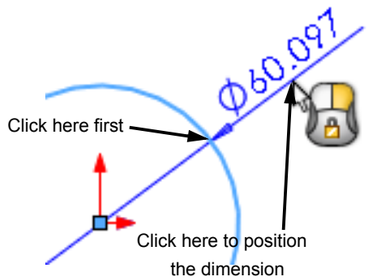


Dimension the sketch

To make the circle the correct size, we will add a diameter dimension to the sketch.

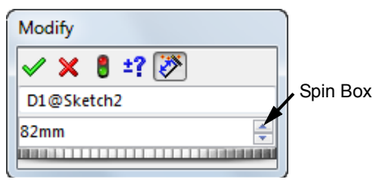
4 Click **Smart Dimension** on the **Sketch** toolbar. The cursor will look like this, indicating that the dimension tool is active.

5 Click on the circle, then move the cursor to the right and up on the screen. The preview dimension and witness lines will be visible. Click to set the dimension location.



6 Input the dimension by typing **82** in the Spin Box. Click to accept the dimension.

Spin boxes are used to input numerical data. They are called spin boxes because the numbers can be spin up or down using the arrows on the right.



中文译文 / CHINESE TRANSLATION

1 单击草图/Sketch 工具栏上的圆/Circle 工具。指针的绘图工具图标下方将显示一个圆，表示圆工具已激活。

2 将指针移到原点上方，直到出现一个橙色圆。绘图工具下方的小黄色图标表示，我们即将使圆心与草图原点重合/Coincident。

3 按住鼠标左键并拖动圆，直到它大约为 32 mm。指针反馈显示的是圆的半径。图中示例反馈为 R = 32.014。

标注草图尺寸/Dimension the sketch

为使圆具有正确大小，需要给草图添加一个直径尺寸。

4 单击草图/Sketch 工具栏上的智能尺寸/Smart Dimension。指针将变为图示形状，表示尺寸工具已激活。

5 单击圆，再将指针向屏幕右上方移动。此时可看到尺寸和尺寸界线的预览。单击以确定尺寸的放置位置。

6 在数值微调框/Spin Box 中键入 82，输入尺寸。单击勾选标记接受该尺寸。

数值微调框/Spin Box 用于输入数值数据。其名称来自右侧箭头：可以利用这些箭头向上或向下调整数值。

图示标注：先单击此处/Click here first；单击此处放置尺寸/Click here to position the dimension；数值微调框/Spin Box。修改/Modify 对话框中输入值为 82 mm；圆上显示的其他数值是标注前的预览值。

ENGLISH ORIGINAL

Sketch Status

Sketches are normally fully defined before creating a feature with them. To be fully defined, the sketch geometry must be geometrically defined and positioned.

- To be geometrically defined, there must be enough dimensions and/or relationships to keep the size and shape of the sketch from changing if we try to drag it.
- To be positioned the sketch must also have dimensions or relationships that keep it from moving.

Sketch Color

The color of the sketch entities shows the status of the individual entity.

Blue - Under defined

Black - Fully defined

Red - Over defined

Extrude

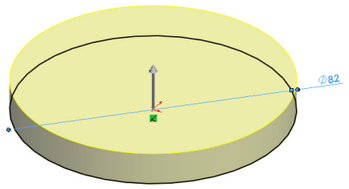
Once the sketch is completed, it can be extruded to create the first feature. There are many options for extruding a sketch including the end conditions, draft and depth of extrusion, which will be discussed in more detail in later lessons. Extrusions take place in a direction normal (perpendicular) to the sketch plane.

Task 3 — Extrude the first feature

Extruding the 2D sketch will produce a 3D solid. In this case, we will make a short cylinder.

1 Select the Features tab on the Command Manager.

Click **Extrude Boss/Base** on the Features toolbar. The model will reorient to the `Isometric` view and show a preview of the extrusion.



2 Preview graphics.

A preview of the feature is shown at the default depth.

A handle appears that can be used to drag the preview to the desired depth. The current depth of the preview can be seen in the PropertyManager.

中文译文 / CHINESE TRANSLATION

草图状态/Sketch Status

通常应先完全定义草图，再用它创建特征。要达到完全定义/Fully defined，既要定义草图几何形状，也要确定其位置。

• 要定义几何形状，需要有足够的尺寸和/或几何关系，使我们尝试拖动草图时，其大小和形状保持不变。

• 要确定位置，草图还必须具有防止其移动的尺寸或几何关系。

草图颜色/Sketch Color

草图实体的颜色表示单个实体的状态：蓝色——欠定义/Under defined；黑色——完全定义/Fully defined；红色——过定义/Over defined。

拉伸/Extrude

草图完成后，即可通过拉伸创建第一个特征。拉伸草图有多种选项，包括终止条件、拔模和拉伸深度；后续课程将详细介绍这些选项。拉伸方向垂直于草图基准面。

任务 3——拉伸第一个特征/Task 3 — Extrude the first feature

拉伸二维草图将生成三维实体。本例中，我们将创建一个短圆柱。

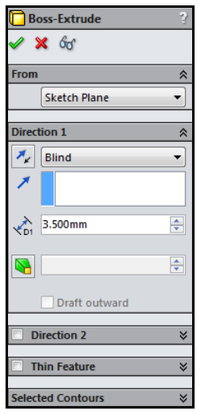
1 选择 CommandManager 上的特征/Features 选项卡。单击特征/Features 工具栏上的拉伸凸台/基体/Extruded Boss/Base。模型会转为等轴测/Isometric 视图，并显示拉伸预览。

2 查看图形预览。特征将按默认深度显示预览。可以拖动出现的控制柄，将预览调整至所需深度。可在 PropertyManager 中查看当前预览深度。

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3 In the PropertyManager, change the settings as shown.

- End Condition = **Blind**
- (Depth) = **3.5mm**



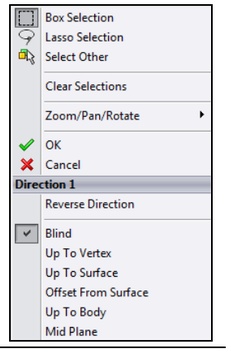
4 Create the extrusion. Click **OK**. The extrusion now becomes a solid and a new feature, `Extrude1` is displayed in the FeatureManager design tree.

TIP:

The **OK** button on the PropertyManager is just one way to complete the command.

A second method is the set of **OK/Cancel** buttons in the confirmation corner of the graphics area.

A third method is the right-mouse shortcut menu that includes **OK**, among other options.



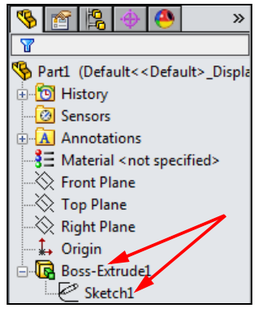
Blind Extrusions

Blind extrusions take the 2D sketch and move it, some specific distance, normal (perpendicular) to the sketch plane.

FeatureManager design tree

This extrusion is the first feature of our part. The FeatureManager design tree shows this feature by type and with a default name Boss-`Extrude1`.

The sketch of the circle (Sketch1) is listed under the feature. It is said to be absorbed by the feature.



中文译文 / CHINESE TRANSLATION

3 在 PropertyManager 中按图示修改设置：终止条件/End Condition = 给定深度/Blind；深度/Depth = 3.5 mm。

4 创建拉伸：单击确定/OK（绿色勾选标记）。拉伸预览随即成为实体，FeatureManager 设计树中显示一个新特征 Extrude1。

给定深度拉伸/Blind Extrusions

给定深度拉伸将二维草图沿垂直于草图基准面的方向移动一个指定距离。

FeatureManager 设计树/FeatureManager design tree

这个拉伸是零件的第一个特征。FeatureManager 设计树按特征类型显示它，并赋予默认名称凸台-拉伸1/Boss-Extrude1。

圆的草图（草图1/Sketch1）列在该特征下方，称为被该特征吸收/absorbed。

提示/TIP：PropertyManager 中的确定/OK 按钮只是完成命令的一种方法。第二种方法是使用图形区域确认角中的确定/取消/OK/Cancel 按钮。第三种方法是使用鼠标右键快捷菜单，其中也包含确定/OK 等选项。

拉伸属性和快捷菜单中的相关标签：从/From；草图基准面/Sketch Plane；方向1/Direction 1；反向/Reverse Direction；成形到一顶点/Up To Vertex；成形到一面/Up To Surface；等距于曲面/Offset From Surface；成形到实体/Up To Body；两侧对称/Mid Plane；向外拔模/Draft outward；方向2/Direction 2；薄壁特征/Thin Feature；所选轮廓/Selected Contours；框选/Box Selection；套索选择/Lasso Selection；选择其他/Select Other；清除选择/Clear Selections；缩放/平移/旋转/Zoom/Pan/Rotate；确定/OK；取消/Cancel。

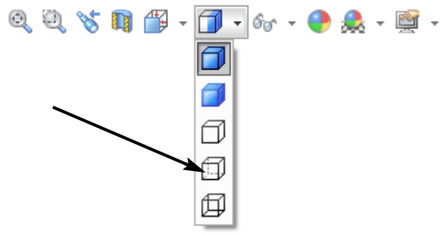
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View Display

The View toolbar provides a quick method to change the way the model is displayed on the screen. It provides one set of tools to Zoom, Pan and Rotate the model view and another to change the way the model is displayed. In most cases, models are created in Shaded view because it most closely resembles the real world.

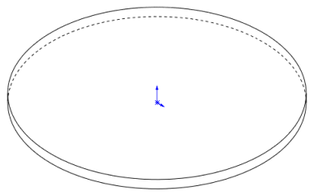


The Head-up View toolbar also provides the same functions. To change the view, click the **Display Style** pull-down list.



Change the display mode. Click **Hidden Lines Visible** on the **View** toolbar.

Hidden Lines Visible allows you to easily select hidden back edges of the part.



Save the Part

Save your work frequently. If you have a computer problem, you may loose everything you did since the last time you saved your work.

5 Click **Save** on the Standard toolbar, or click **File, Save.**

The **Save As** dialog box appears.

6 Type `Binding Anchor` for the filename.

中文译文 / CHINESE TRANSLATION

视图显示/View Display

视图/View 工具栏可以快速改变模型在屏幕上的显示方式。它提供一组用于缩放/Zoom、平移/Pan 和旋转/Rotate 模型视图的工具，以及另一组用于更改模型显示样式的工具。大多数情况下，创建模型时使用上色/Shaded 视图，因为它最接近真实物体的外观。

前导视图工具栏/Heads-up View toolbar 也提供相同功能。若要更改视图显示样式，单击显示样式/Display Style 下拉列表。

更改显示模式：单击视图/View 工具栏上的隐藏线可见/Hidden Lines Visible。显示隐藏线便于选取零件背面被遮挡的边。

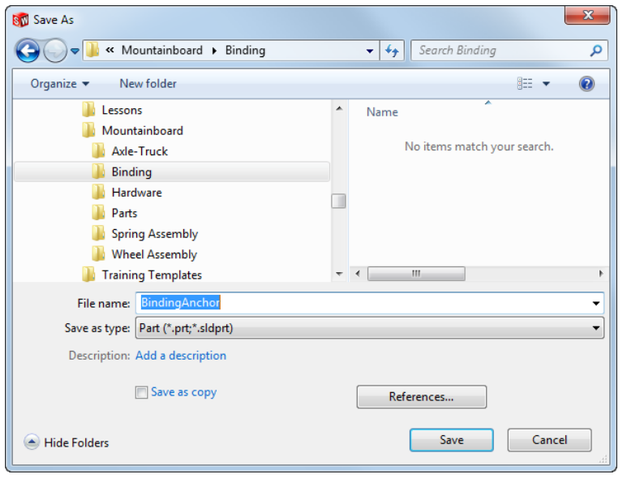
保存零件/Save the Part

请经常保存工作。如果计算机出现问题，上次保存之后所做的全部工作都可能丢失。

5 单击标准/Standard 工具栏上的保存/Save，或依次单击文件/File、保存/Save。出现另存为/Save As 对话框。

6 输入文件名 Binding Anchor。

ENGLISH ORIGINAL



7 Save the file to the folder `Binding` found under `SolidWorks Curriculum` and `Lessons_<year>\Mountainboard Design Project\Mountainboard`.

8 Click **Save**.

The `sldprt` extension is automatically added to the filename.

The file is saved to the current folder. You can use the Windows browse button to change to a different folder.

Note: All the files we create of the Mountainboard project should be saved in the appropriate folder under the folder …`\SolidWorks Curriculum` and `Lessons_<year>\Mountainboard Design Project`.

Changing views

The Standard Views toolbar or the Heads-Up View toolbar makes it easy to change your view of the model by simply clicking on the view you would like to see.

中文译文 / CHINESE TRANSLATION

7 将文件保存到 SolidWorks Curriculum and Lessons_<year>\Mountainboard Design Project\Mountainboard 下的 Binding 文件夹。

8 单击保存/Save。文件名会自动添加 .sldprt 扩展名。文件将保存到当前文件夹；可使用 Windows 的浏览按钮改为其他文件夹。

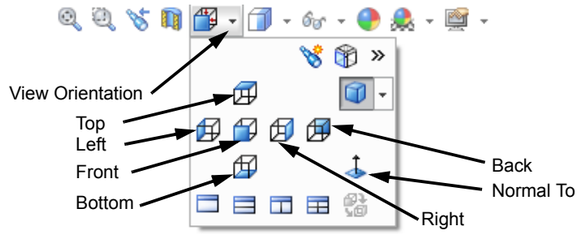
切换视图/Changing views

标准视图/Standard Views 工具栏或前导视图工具栏/Heads-up View toolbar 可让你方便地切换模型视图，只需单击要查看的视图即可。

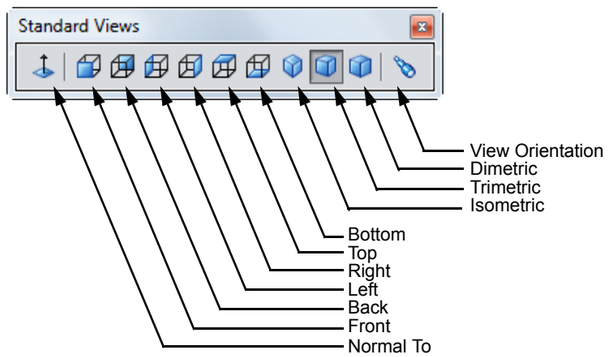
注意/Note：山地滑板/Mountainboard 项目中创建的所有文件，都应保存到 …\SolidWorks Curriculum and Lessons_<year>\Mountainboard Design Project 下的相应文件夹。

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Heads-up View Toolbar



Standard Views Toolbar



Mouse Gestures

Mouse Gestures can also be used to access the different views. The tools available through Mouse Gestures can be customized to show either four or eight tools. The tools available will also depend on whether you are in a sketch, part, assembly or drawing. Shown are the default Mouse Gestures for a part.



9 Change the view of the model to the **Bottom** view. Click on either the **Standard Views** or **Heads-up View** toolbar.

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前导视图工具栏/Heads-up View Toolbar

图示命令：视图定向/View Orientation；上视/Top；左视/Left；前视/Front；下视/Bottom；后视/Back；正视于/Normal To；右视/Right。

标准视图工具栏/Standard Views Toolbar

图示命令：正视于/Normal To；前视/Front；后视/Back；左视/Left；右视/Right；上视/Top；下视/Bottom；等轴测/Isometric；三轴测/Trimetric；二等角/Dimetric；视图定向/View Orientation。

鼠标笔势/Mouse Gestures

也可使用鼠标笔势/Mouse Gestures 调出不同视图。可以自定义鼠标笔势中可用的工具，显示四个或八个工具。可用工具也取决于当前处于草图、零件、装配体还是工程图环境。图中显示的是零件环境的默认鼠标笔势。

9 将模型视图切换为下视/Bottom 视图。单击标准视图/Standard Views 工具栏或前导视图工具栏/Heads-up View toolbar 上的下视/Bottom 按钮。

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Task 4 — Add a second feature

The second feature will be another cylinder, slightly smaller than the first.

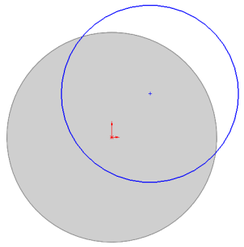
1 Change the display mode back to shaded. Click **Shaded With Edges** on the **View** toolbar.

2 Select the bottom face of the cylinder. It will turn blue to show that it is selected.

3 Start a new sketch by clicking the **Sketch** on the Sketch toolbar or pop up toolbar.

4 Select **Circle** on the Sketch toolbar.

5 Draw a circle, slightly smaller than the size of the cylinder. It does not have to be centered on the cylinder. The circle is blue, indicating that the sketch is Under Defined.



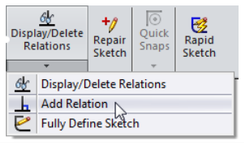
Task 5 — Adding sketch relationships

Sketch relationships are used to force a behavior on a sketch element to capture design intent. Some are automatic, others can be added as needed.

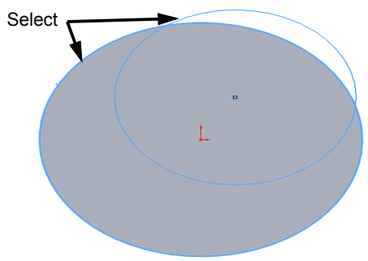
When adding relationships, only those relationships that are appropriate for the sketch entities selected will be shown in the PropertyManager.

1 Click **Add Relation** on the Sketch toolbar. **Add Relation** will appear in the PropertyManager.

Note: When using the Command Manager, click **Display/Delete Relations**, then select **Add Relations** from the dropdown list.



2 Select the circle and the edge of the cylinder.



中文译文 / CHINESE TRANSLATION

任务 4—添加第二个特征 / Task 4—Add a second feature

第二个特征也是一个圆柱体，比第一个稍小。

1. 将显示模式改回着色。单击视图 / View 工具栏上的带边线上色 / Shaded With Edges。

2. 选择圆柱体的底面。该面会变为蓝色，表示已被选中。

3. 单击草图 / Sketch 工具栏或弹出工具栏上的草图 / Sketch，开始绘制新草图。

4. 在草图 / Sketch 工具栏上选择圆 / Circle。

5. 绘制一个比圆柱体稍小的圆。此时不必让圆与圆柱体同心。圆呈蓝色，表示草图处于欠定义 / Under Defined 状态。

任务 5—添加草图几何关系 / Task 5—Adding sketch relationships

草图几何关系使草图元素遵循特定的行为，从而表达设计意图。有些几何关系会自动添加，其他关系可按需添加。

添加几何关系时，属性管理器 / PropertyManager 仅显示适用于所选草图实体的关系。

1. 单击草图 / Sketch 工具栏上的添加几何关系 / Add Relation。属性管理器中将出现添加几何关系面板。

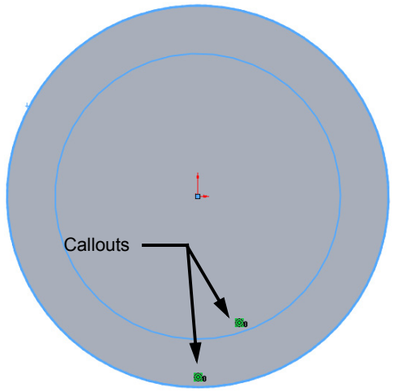
2. 选择草图圆和圆柱体的圆边线。

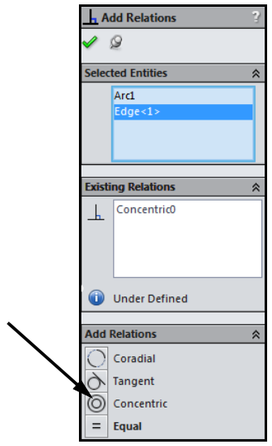
注：使用命令管理器 / CommandManager 时，单击显示/删除几何关系 / Display/Delete Relations，然后从下拉列表中选择添加几何关系 / Add Relations。

图中标注：选择 / Select。

ENGLISH ORIGINAL

3 Click **Concentric** in the **Add Relations** box. Then click **OK**. The circle will move to a position where it is centered on the cylinder. Callouts will show on the sketch to show the relationships.





Callouts

Callouts provide a display of existing conditions. The relationship callouts show which relationships exist and between which sketch entities.

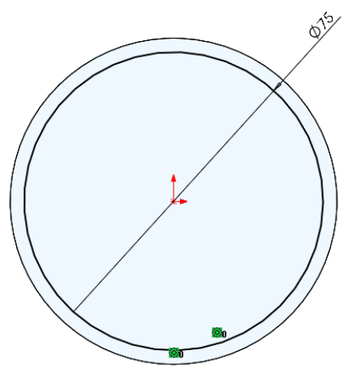
Task 6 — Dimension the circle

The concentric relationship defines the position of the circle, but it is still blue (under defined) because it doesn't have a dimension for its diameter.

1 Click **Smart Dimension** on the Sketch toolbar.

2 Click on the circle, move the cursor to the right the click again to set the dimension position.

3 Type **75** for the value. Click . The circle will now be black to show that it is fully defined.



Status Bar

The status bar, located at the bottom right of the graphics window will also show the state of the sketch. Fully Defined

Change the viewpoint

When creating the first feature, our viewpoint was automatically changed to the `Isometric` view. After the first feature, we must change the view to best see the preview of the new feature.

中文译文 / CHINESE TRANSLATION

3. 在添加几何关系 / Add Relations 框中单击同心 / Concentric，然后单击确定 / OK。草图圆将移动到与圆柱体同心的位置。草图中出现几何关系标注。

标注 / Callouts

标注显示当前已有的条件。几何关系标注显示存在哪些关系，以及关系关联了哪些草图实体。

任务 6—标注圆的尺寸 / Task 6—Dimension the circle

同心关系确定了圆的位置，但圆仍呈蓝色，即欠定义，因为还没有直径尺寸。

1. 单击草图 / Sketch 工具栏上的智能尺寸 / Smart Dimension。

2. 单击圆，将光标移到右侧，再次单击以放置尺寸。

3. 输入数值 75，单击确认按钮。圆现在变为黑色，表示完全定义 / Fully Defined。

状态栏 / Status Bar

位于图形窗口右下角的状态栏也显示草图的定义状态。

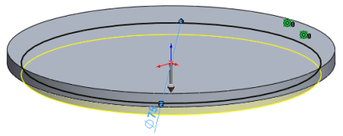
改变观察方向 / Change the viewpoint

创建第一个特征时，观察方向自动切换为等轴测 / Isometric 视图。第一个特征创建之后，需要自行调整观察方向，以便清楚观察新特征的预览。

图中标注：标注 / Callouts。

Task 7 — Extrude the second feature

1. Change the viewpoint to `Dimetric` by clicking on the Views Orientation toolbar.
2. Click **Extrude** on the Features toolbar.
3. Select **Blind** for the type of extrusion.
4. Type **3mm** for the depth. Check the preview shown in yellow. It shows that material will be added to the bottom of the first cylinder.
5. Click .



Task 8 — Cut a recess in the top of the part

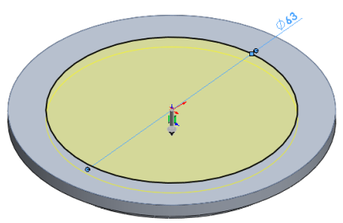
Material needs to be removed from the top of this part to:

- ❑ Reduce weight. Each part must be designed to be as light as possible so that the assembled mountainboard is not too heavy to be carried.
- ❑ Lower the tops of the screws used to bolt this part to the deck. This will reduce the chance of anything (pants leg, shoe laces, etc.) getting caught on the screw heads.

Removing material by extruding a cut

Material can be added or removed by extrusion. The process to add or remove material works the same in that you start by creating a 2D sketch. That sketch is then moved normal to the sketch plane. If you are creating a boss, the enclosed volume is added to the part. If you are creating a cut, the enclosed volume is removed from the part.

1. Orient the part to the `Top` view by clicking on the **Views Orientation** toolbar.
2. Select the top face of the model, and click **Sketch** to start a new sketch.
3. Click **Circle** on the Sketch toolbar.
4. Draw a circle from the center of the top face. Make its radius about **30mm**.
5. Dimension the circle to be **63mm** in diameter.
6. Reorient the model to the `Isometric` view.
7. Click **Extruded Cut** to use the circle to cut away some material.
8. Check the preview, by default, cuts go into the existing part.
9. Type **3mm** for the depth.



任务 7—拉伸第二个特征 / Task 7—Extrude the second feature

1. 单击视图方向 / View Orientation 工具栏上的相应按钮，将观察方向改为二等角轴测 / Dimetric。
2. 单击特征 / Features 工具栏上的拉伸 / Extrude。
3. 将拉伸类型设为给定深度 / Blind。
4. 输入深度 3 mm。检查黄色预览；它应显示在第一个圆柱体的底部添加材料。
5. 单击确认按钮。

任务 8—在零件顶部切除凹槽 / Task 8—Cut a recess in the top of the part

需要从零件顶部去除材料，以实现下列目的：

• 减轻重量。每个零件都应尽量轻，以免装配完成的山地滑板过重、不便携带。

• 降低用于将本零件固定到板面的螺钉头高度，减少裤脚、鞋带等被螺钉头勾住的可能性。

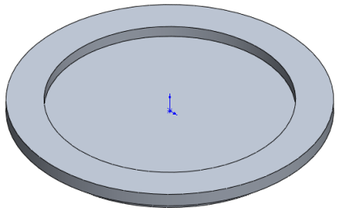
通过拉伸切除去除材料 / Removing material by extruding a cut

拉伸既可以添加材料，也可以去除材料。两者的过程相同：先创建二维草图，再沿垂直于草图平面的方向移动截面。创建凸台时，将所围成的体积添加到零件；创建切除时，从零件中去除该体积。

1. 单击视图方向 / View Orientation 工具栏上的相应按钮，将零件转到上视 / Top 视图。
2. 选择模型顶面，单击草图 / Sketch，开始新草图。
3. 单击草图 / Sketch 工具栏上的圆 / Circle。
4. 以顶面的中心为圆心绘制一个圆，初始半径约为 30 mm。
5. 将圆的直径标注为 63 mm。
6. 将模型转到等轴测 / Isometric 视图。
7. 单击拉伸切除 / Extruded Cut，利用这个圆切除材料。
8. 检查预览。切除默认朝现有零件内部进行。
9. 输入深度 3 mm。

ENGLISH ORIGINAL

10 Click ✔. The **Cut Extrude** command removed a cylinder shaped volume from the part.



Calculating the weight of the part

In any design, it is important to keep track of the weight of each individual part. In the case of the Mountainboard, if the individual parts become too heavy, the total weight of the board may exceed a reasonable weight to be carried up the hill.

The weight of the part can be calculated by multiplying the volume of the part by the density of the material.

- Weight = Volume x Density

Calculate the volume

The total volume is the sum of the volumes created by the two extrudes minus the volume of the cut.

- Total Volume = Volume of each of the two extruded cylinders - volume of the extruded cut
- Volume of a cylinder = Area of the circle times the cylinder height = Pi times the diameter squared divided by 4, times the cylinder height = (Pi * D^2/4) h
- Total Volume = (3.14 * 82^2 / 4)(3.5) + (3.14 * 75^2 / 4)(3) - (3.14 * 63^2 / 4)(3) = 18,483.56 + 13,253.60 - 9,351.74 = **22,385.42 cubic millimeters**

Find the density

The density of engineering materials can be found in many ways. There are numerous engineering handbooks or several sites on the internet. One such site is MatWeb (www.matweb.com).

The `Binding Anchor` will be made from 2015 Aluminum. Using MatWeb, the density for 2015 Aluminum is 2.8 g/cc. There are 1000 cubic mm in 1 cc, so the density would be:

2.8 g/cc x .001 cc/mm = **.0028 g/mm^3**

Calculate the weight

Weight = Volume x Density = 22,385.42mm^3 x .0028 g/mm^3 = **62.68 grams (2.21 oz)**.

中文译文 / CHINESE TRANSLATION

10. 单击确认按钮。拉伸切除 / Cut Extrude 命令从零件中去除了一个圆柱形体积。

计算零件的重量 / Calculating the weight of the part

在任何设计中，都应关注各个零件的重量。对于山地滑板，如果各零件过重，整块滑板的总重量就可能超过适合背上山的合理范围。

零件的重量可用其体积乘以材料密度计算。

• 重量 = 体积 × 密度 / Weight = Volume × Density

计算体积 / Calculate the volume

总体积等于两个拉伸生成的体积之和，减去切除的体积。

• 总体积 = 两个拉伸圆柱体的体积之和 − 拉伸切除的体积。

• 圆柱体体积 = 圆面积 × 圆柱高度 = π × 直径平方 ÷ 4 × 圆柱高度 = (πD^2/4)h。

• 总体积 = (3.14 × 82^2 / 4) × 3.5 + (3.14 × 75^2 / 4) × 3 − (3.14 × 63^2 / 4) × 3 = 18,483.56 + 13,253.60 − 9,351.74 = 22,385.42 mm^3。

查找密度 / Find the density

工程材料的密度可通过多种途径查得，例如工程手册或互联网上的资料网站，其中一个网站是 MatWeb（www.matweb.com）。

固定器锚座 / Binding Anchor 将采用 2015 铝合金 / 2015 Aluminum 制造。根据 MatWeb，2015 铝合金的密度为 2.8 g/cc。1 cc 等于 1,000 mm^3，因此密度换算如下：

2.8 g/cc × 0.001 cc/mm^3 = 0.0028 g/mm^3。

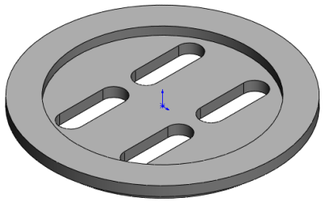
计算重量 / Calculate the weight

重量 = 体积 × 密度 = 22,385.42 mm^3 × 0.0028 g/mm^3 = 62.68 g（2.21 oz，盎司）。

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Task 9 — Create the screw slots

To make the position of the binding adjustable, the binding anchor will have four slots. These will allow the position of the bindings to be moved along the centerline of the deck.



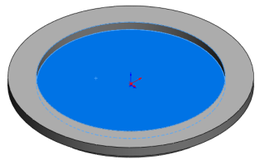
The slots are symmetrical, so we will use a function called mirroring to make sure the slots always remain symmetrical if we later need to change their size.

Create a sketch

1 Select the face of the part created by the cut.

2 **Open** a sketch by clicking on the **Sketch** toolbar.

3 Change to the Top view by clicking on the **View Orientation** toolbar.



The Sketch Mirror tool

Mirror Entities and **Dynamic Mirror Entities** create symmetric relationships between sketch entities about a centerline. The **Dynamic Mirror Entities** command can be used *while sketching* and the **Mirror Entities** *after sketching* with the same final results.

Lines and Centerlines

The Line tool draws straight lines. If the line is vertical, the cursor will show a yellow callout to indicate that a **Vertical** relationship will be added. If the line is horizontal, the cursor will show an to indicate that a **Horizontal** relationship will be added.

Centerlines are construction geometry. They are used to position other entities but do not result in features.

While you are sketching, the callouts will be yellow , indicating which relationship will be added when you release the mouse button. The green callouts show relationships that have been added .

Note: The color of the callouts indicating existing relationships, green in this case, can also be cyan. Their color depends on the color scheme chosen in the SolidWorks Options.

Create a centerline

4 Click **Centerline** on the Sketch toolbar.

中文译文 / CHINESE TRANSLATION

任务 9—创建螺钉长圆槽 / Task 9—Create the screw slots

为使固定器的位置可调，固定器锚座 / Binding Anchor 上将设置四个槽。这些槽允许固定器沿板面中心线方向移动。

这些槽相互对称，因此将使用镜向 / Mirroring 功能，确保以后修改尺寸时它们始终保持对称。

创建草图 / Create a sketch

1. 选择由上一切除特征形成的面。

2. 单击草图 / Sketch 工具栏上的相应按钮，开始绘制草图。

3. 单击视图方向 / View Orientation 工具栏上的相应按钮，将视图切换为上视 / Top。

草图镜向工具 / The Sketch Mirror tool

镜向实体 / Mirror Entities 和动态镜向实体 / Dynamic Mirror Entities 均可在草图实体之间建立关于中心线的对称关系。动态镜向实体可在绘图过程中使用，镜向实体可在绘图完成后使用；两者最终结果相同。

直线与中心线 / Lines and Centerlines

直线 / Line 工具绘制直线。当直线竖直时，光标显示黄色标注，表示将添加竖直 / Vertical 关系；当直线水平时，光标显示相应标注，表示将添加水平 / Horizontal 关系。

中心线 / Centerlines 属于构造几何线。它们用于定位其他实体，但本身不生成实体特征。

绘图过程中，黄色标注表示释放鼠标按钮时将添加的几何关系；绿色标注表示已经添加的几何关系。

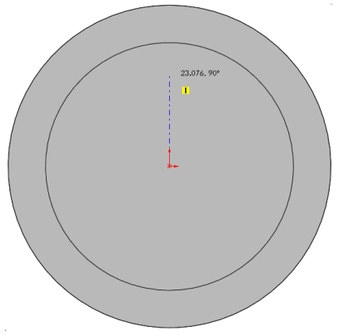
创建中心线 / Create a centerline

4. 单击草图 / Sketch 工具栏上的中心线 / Centerline。

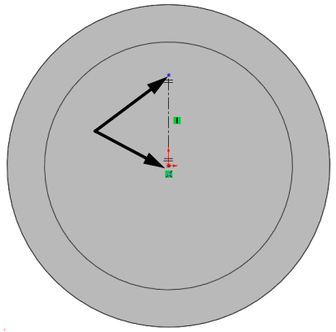
注：表示已有几何关系的标注，本例为绿色，也可能为青色；其颜色取决于 SOLIDWORKS 选项 / Options 中选择的配色方案。

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5 Sketch a vertical **Centerline** from the sketch origin. The length is not important. Make sure the cursor displays a |, indicating a **Vertical** relationship will be added.

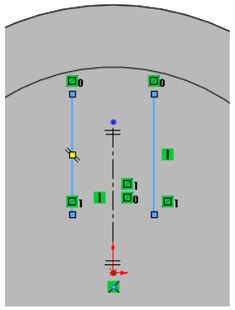


6 Click **Dynamic Mirror Entities** on the **Sketch** toolbar. A pair of parallel marks will appear at each end of the centerline to show that we are in the mirror mode.



7 Sketch a vertical line to one side of the centerline. As soon as you finish drawing the line, a mirror image will be drawn automatically on the other side of the centerline.

The callouts will show that a symmetric relationship has been added between the endpoints of each line.



Arcs

There are three tools provided to create arcs:

Tangent Arc — Adds a tangent relationship to the entity it is sketched from.

Center Point Arc — Defined by a center point and a radius.

3 Point Arc — Defined by two endpoints and a radius.

The choice of arc tools depends on the geometry that needs to be created.

Add an Arc

The sketch of the slot is composed of two straight lines and two arcs. The arcs must be tangent to both lines.

中文译文 / CHINESE TRANSLATION

5. 从草图原点开始绘制一条竖直中心线 / Centerline。长度不重要。确认光标显示竖直关系标志，表示将添加竖直 / Vertical 关系。

6. 单击草图 / Sketch 工具栏上的动态镜向实体 / Dynamic Mirror Entities。中心线两端各出现一对平行标记，表示已进入镜向模式。

7. 在中心线一侧绘制一条竖直线。画完后，系统立即在中心线另一侧自动生成其镜像。标注会显示，两条直线的对应端点之间已添加对称 / Symmetric 关系。

圆弧 / Arcs

软件提供三种绘制圆弧的工具：

• 切线弧 / Tangent Arc：从现有实体开始绘制圆弧，并自动与该实体建立相切 / Tangent 关系。

• 圆心/起/终点画弧 / Center Point Arc：通过圆心和半径定义圆弧。

• 3 点圆弧 / 3 Point Arc：通过两个端点和半径定义圆弧。

选择哪种圆弧工具，取决于需要创建的几何形状。

添加圆弧 / Add an Arc

槽的草图由两条直线和两段圆弧组成；两段圆弧必须分别与两条直线相切。

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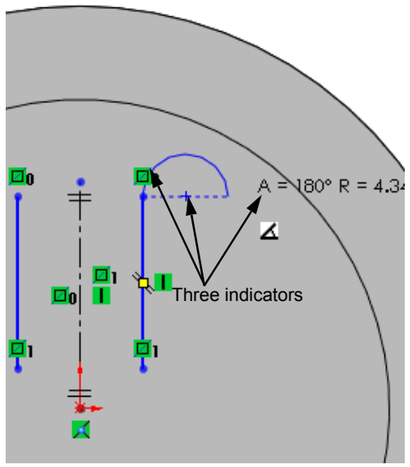
8 Click **Tangent Arc** the **Sketch** toolbar.

9 Place the cursor over the end of the right vertical line and drag an arc up and around to the right until you get cursor feedback showing you have gone 180 degrees.

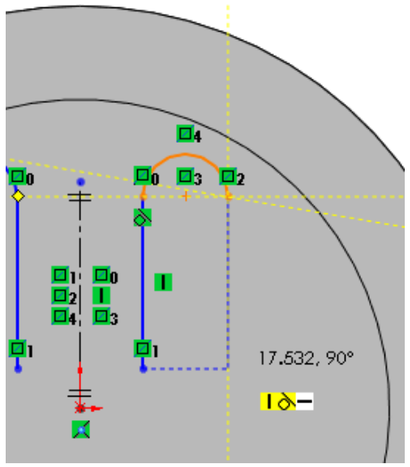
There will be three indicators that you have gone 180 degrees:

- A blue dashed line (inference line) from the center of the arc
- The angle symbol under the drawing tool
- The arc degree feedback (A=180)

When you release the mouse button, a second symmetric arc will be drawn automatically.



10 Draw another vertical line, from the end of the arc, vertically downward until you get an blue inference line from the bottom end of the first line.



中文译文 / CHINESE TRANSLATION

8. 单击草图 / Sketch 工具栏上的切线弧 / Tangent Arc。

9. 将光标放在右侧竖直线的端点上，向上并向右绕转拖出圆弧，直到光标反馈显示圆弧达到 180°。

达到 180° 时有三种指示：

• 从圆弧中心延伸出的蓝色虚线，即推理线 / Inference line。

• 绘图工具光标下方的角度符号。

• 圆弧角度反馈 A = 180。

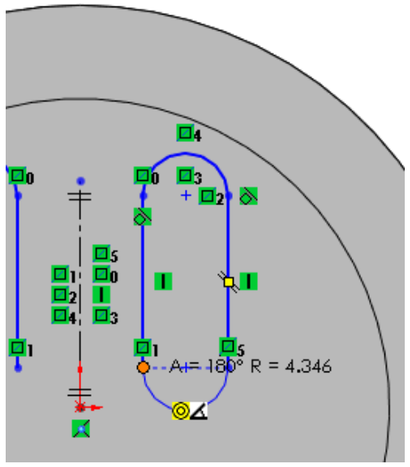
释放鼠标按钮后，系统自动生成第二段对称圆弧。

10. 从该圆弧的另一端点开始，竖直向下绘制另一条直线，直到出现从第一条直线底端延伸过来的蓝色推理线。

图中标注：三种指示 / Three indicators。

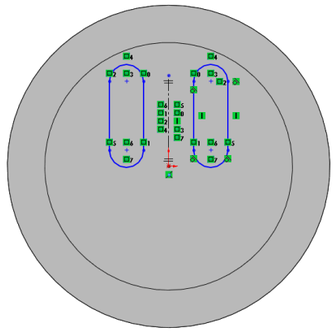
11 Finish the sketch with another **Tangent Arc**.

12 Turn off the sketch **Dynamic Mirror Entities** tool by clicking on the Sketch toolbar.



Review the progress

With sketch mirroring turned on, each entity we drew had a mirrored entity drawn on the other side of the centerline. The symmetric relationships added by the mirror tool will make these sketch elements retain the symmetry we desire.



Callouts show that the arcs are tangent to the two lines it connects to and symmetric to the other arcs and centerlines . The numbers next to each symmetric relationship show the pairs of symmetric elements.

With all the symmetric relationships, the number of callouts displayed may make the sketch elements hard to see, so we can turn them off.

Turn off the callouts

To toggle off the callout display of existing relationships, click **View, Sketch Relations** from the menu. This command is a toggle that turns the callouts on or off.

13 Click **View, Sketch Relations**.

14 Turn off the callouts by clicking **View, Sketch Relations**.

Mirror after sketching

Mirrored entities can also be created after creating sketch entities. To mirror after sketching, select the centerline about which you want to mirror and all the entities you want to mirror.

To mirror after sketching, click **Mirror Entities** .

11. 用另一段切线弧 / Tangent Arc 闭合草图。

12. 单击草图 / Sketch 工具栏上的动态镜向实体 / Dynamic Mirror Entities 按钮，关闭该工具。

检查当前进度 / Review the progress

启用草图镜向时，每画一个实体，中心线另一侧便自动生成其镜像。镜向工具添加的对称关系使草图元素保持所需的对称性。

标注显示，圆弧与其连接的两条直线相切 / Tangent，并且与其他圆弧及中心线保持对称 / Symmetric。每个对称关系旁的编号表示一对对应的对称元素。

由于对称关系很多，显示的标注可能使草图实体难以看清，因此可以关闭标注显示。

关闭标注 / Turn off the callouts

要切换已有几何关系标注的显示状态，选择菜单中的视图 / View > 草图几何关系 / Sketch Relations。此命令用于开启或关闭标注。

13. 单击视图 / View > 草图几何关系 / Sketch Relations。

14. 单击视图 / View > 草图几何关系 / Sketch Relations，关闭标注。

绘图后镜向 / Mirror after sketching

也可以在创建草图实体之后再生成镜向实体。操作时，选择作为镜向轴的中心线，以及所有需要镜向的实体。

然后单击镜向实体 / Mirror Entities。

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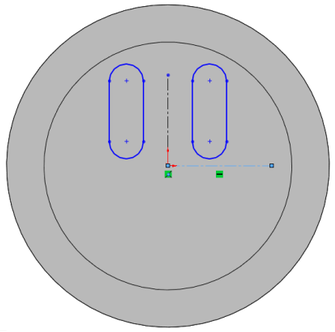
Task 10 — Mirror the two slot sketches

To complete the pattern of slots, mirror the two slots across a horizontal centerline.

1 Sketch a horizontal centerline from the origin to the right.

2 Click **Mirror Entities** on the Sketch toolbar.

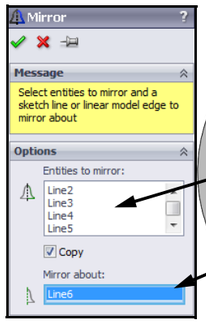
3 Turn off the **Centerline** tool by clicking on the tool again.

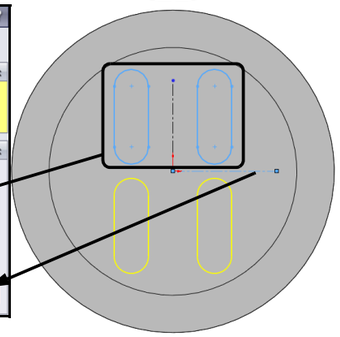


4 Select the four lines and four arcs as the **Entities to mirror**. Select the horizontal centerline for the entity to **Mirror about**.

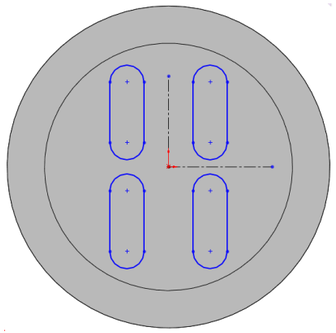
Make sure you *do not* have the vertical centerline selected.

Click .





5 We now have a sketch that will cut the four slots.



中文译文 / CHINESE TRANSLATION

任务 10—镜向两条槽的草图 / Task 10—Mirror the two slot sketches

为完成槽的排列，将两条槽关于水平中心线进行镜向。

1. 从原点向右绘制一条水平中心线 / Centerline。

2. 单击草图 / Sketch 工具栏上的镜向实体 / Mirror Entities。

3. 再次单击中心线 / Centerline 工具按钮以关闭该工具。

4. 在要镜向的实体 / Entities to mirror 中选择四条直线和四段圆弧。在镜向点 / Mirror about 中选择水平中心线。确保没有选中竖直中心线。单击确认按钮。

5. 此时得到的草图将用于切除四条槽。

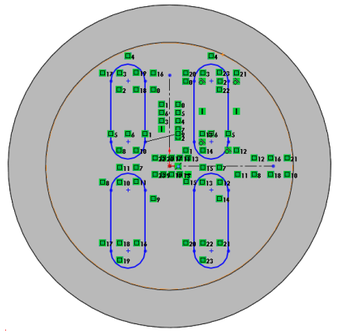
图中镜向 / Mirror 属性管理器的文字：选择要镜向的实体，以及用作镜向轴的草图直线或模型边线 / Select entities to mirror and a sketch line or model edge to mirror about；选项 / Options；要镜向的实体 / Entities to mirror；复制 / Copy；镜向点 / Mirror about。

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View Relationships

6 Click **View, Sketch Relations**.

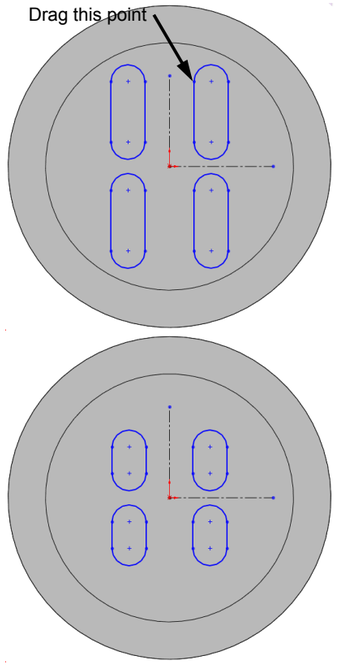
The callouts show that the all the arcs are tangent to the two lines they connect to and symmetric to the other arcs and centerlines . The numbers next to each symmetric relationship show the pairs of symmetric elements.



7 With this many symmetric pairs, the number of callouts can make it difficult to see the sketch. Turn off the callouts by clicking **View, Sketch Relations**.

Test the relationships

All four slots sketches should have symmetric relationships. Anything done to one slot should be mirrored into the other sketches.



8 Drag the point shown. All four slots should change shape together.

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查看几何关系 / View Relationships

6. 单击视图 / View > 草图几何关系 / Sketch Relations。

标注显示，所有圆弧均与其连接的两条直线相切 / Tangent，并且与其他圆弧及中心线保持对称 / Symmetric。每个对称关系旁的编号表示一对对应的对称元素。

7. 对称实体对较多时，标注会使草图难以看清。单击视图 / View > 草图几何关系 / Sketch Relations，关闭标注。

检验几何关系 / Test the relationships

四条槽的草图都应具有对称关系。对其中一条槽所作的任何改动，都应在其他槽中得到相应的镜向变化。

8. 拖动图示的点。四条槽应一起改变形状。

图中标注：拖动此点 / Drag this point。

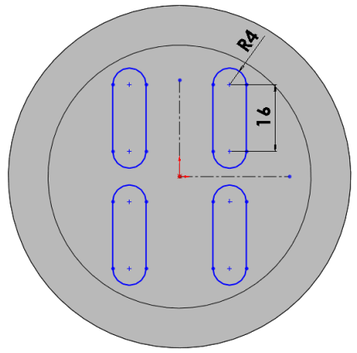
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Task 11 — Add dimensions

To fully define the sketch, we must add dimensions to position the slots and define their size. Even though we only drew one of the slots, the dimensions can be on any of the slots.

1 Add dimensions to the upper right slot as shown.

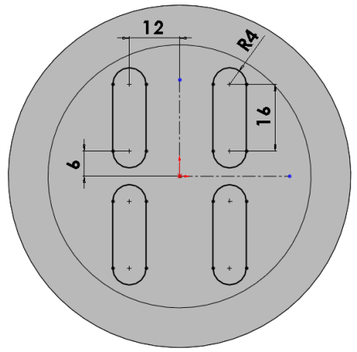
To add the **16mm** dimension, select the two arcs, not the vertical line.



2 Add dimensions as shown to the upper left slot to position it.

Both of these dimensions go from the lower arc to one of the centerlines.

3 Fully defined. The sketch geometry should now be all black, showing that the sketch is fully defined.



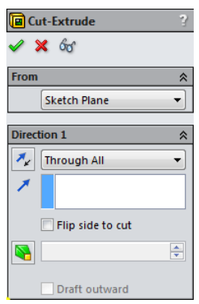
Task 12 — Create the cuts

The four slots must cut completely through the `Binding Anchor`. When we create the cut, it must be done so that if we need to change the thickness of the material later in the design process, the slots do not have to be redone.

1 Click **Insert, Cut, Extrude** from the menu.

2 Click on the **View** toolbar to change the view to `Isometric`.

3 From the list in the PropertyManager select **Through All**.



Through All

The end condition **Through All** will make the cut go through all the geometry. If, when we later analyze the part for strength we determine that it needs to thicker, we will not have to redo the slots because they will go through the entire part, no matter how thick it is.

中文译文 / CHINESE TRANSLATION

任务 11 — 添加尺寸 / Add dimensions

为了完全定义草图，必须添加尺寸来确定槽的位置和大小。虽然我们只绘制了其中一个槽，但这些尺寸可以标注在任意一个槽上。

1. 按图示为右上方的槽添加尺寸。标注 16 mm 尺寸时，应选择两段圆弧，而不是竖直直线。图中圆弧半径为 R4，两个端部圆弧的圆心间距为 16 mm。

2. 按图示为左上方的槽添加尺寸，以确定其位置。这两个定位尺寸均从下端圆弧标注至其中一条中心线：下端圆弧圆心到竖直中心线的水平距离为 12 mm，到水平中心线的竖直距离为 6 mm。

3. 完全定义 / Fully defined：此时草图几何体应全部变为黑色，表明草图已完全定义。

任务 12 — 创建切除 / Create the cuts

四个槽必须完全切穿固定器锚盘 / Binding Anchor。创建切除时，应保证以后在设计过程中需要更改材料厚度时，不必重新创建这些槽。

1. 从菜单选择插入 / Insert → 切除 / Cut → 拉伸 / Extrude。

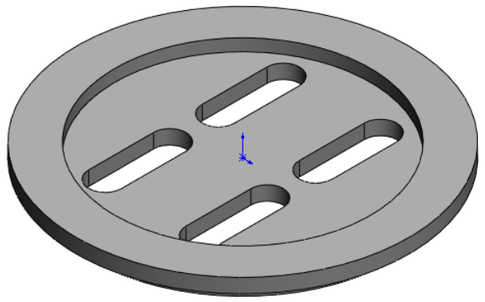
2. 单击视图 / View 工具栏中的等轴测 / Isometric 按钮，将视图改为等轴测。

3. 从属性管理器 / PropertyManager 的列表中选择完全贯穿 / Through All。

完全贯穿 / Through All：该终止条件使切除贯穿全部几何体。如果以后分析零件强度时确定需要增加厚度，就不必重新创建这些槽，因为无论零件有多厚，槽都会贯穿整个零件。

ENGLISH ORIGINAL

4 Complete the cut. Click ✔.



Renaming Features

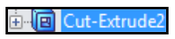
All the features shown in the FeatureManager design tree can be renamed. Renaming features can make them easier to locate as the parts become more complex.

There are three methods to rename features:

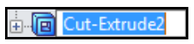
- **Click-Pause-Click. Click** on the feature name, **Pause**, **Click** the name again, type the new name.
- Click on the feature name, press **F2**, type the new name.
- Right-click the feature name and select **Properties**. Change the name in the Properties dialog box.

Task 13 — Rename the slots

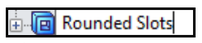
1 In the FeatureManager design tree, click once on `Cut-Extrude2`, this is the slots we just created.



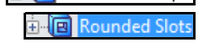
2 Press **F2**. The feature name now has a box around it and a flashing cursor.



3 Type `Rounded Slots` for the new name.



4 The feature has now been renamed to something more descriptive.



Notice that the feature's icon does not change. The  shows that this feature is a **Cut-Extrude**.

中文译文 / CHINESE TRANSLATION

4. 完成切除：单击确定 / OK（绿色对勾）。

重命名特征 / Renaming Features

FeatureManager 设计树 / FeatureManager design tree 中显示的所有特征都可以重命名。随着零件变得更加复杂，为特征重命名可以使其更容易查找。

重命名特征有三种方法：① 单击—暂停—再单击：单击特征名称，稍停片刻，再次单击该名称，然后输入新名称。② 单击特征名称，按 F2，然后输入新名称。③ 右键单击特征名称并选择属性 / Properties，在属性 / Properties 对话框中更改名称。

任务 13 — 重命名槽 / Rename the slots

1. 在 FeatureManager 设计树中单击一次切除-拉伸2 / Cut-Extrude2；这就是刚创建的槽切除特征。

2. 按 F2。特征名称周围会出现文本框，并显示闪烁的光标。

3. 输入 Rounded Slots（圆头槽），作为新名称。

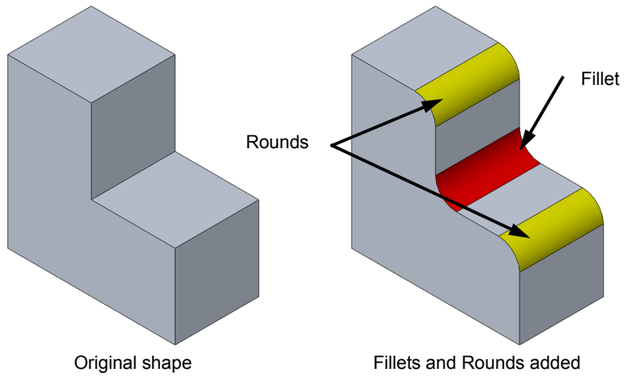
4. 该特征现在已改为更具描述性的名称。注意，特征图标不会改变；图中的图标仍表明这是一个拉伸切除 / Cut-Extrude 特征。

ENGLISH ORIGINAL

Filleting

Filleting refers to both fillets and rounds. The distinction is made by the geometric conditions, not the command itself. Fillets are created on selected edges. Those edges can be selected in several ways.

Both fillets (adding volume), shown in red, and rounds (removing volume), shown in yellow, are created with this command. The orientation of the edge or face determines which is used.

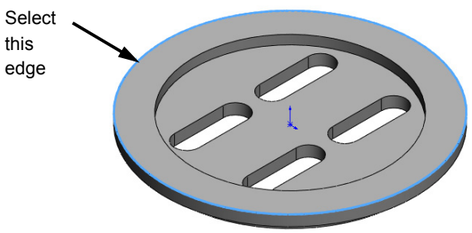


Task 14 — Round an outside corner

All the existing edges in this model are sharp. To meet our design intent, all the exposed edges need to be rounded.

1 Click **Fillet** on the **Features** toolbar.

2 Select the edge shown.



中文译文 / CHINESE TRANSLATION

圆角 / Filleting

Filleting 同时指内圆角 / Fillets 和外圆角 / Rounds。两者由几何条件区分，而不是由命令本身区分。圆角创建在选定的边线上，而选择这些边线有多种方法。

使用同一个命令，既可以创建图中红色的内圆角（增加体积），也可以创建图中黄色的外圆角（去除体积）。边线或面的朝向决定采用哪一种。

图示标注：Original shape＝原始形状；Fillets and Rounds added＝添加内圆角与外圆角后；Fillet＝内圆角；Rounds＝外圆角。

任务 14 — 修圆一个外侧棱边 / Round an outside corner

当前模型上的所有边线均为锐边。为了满足设计意图，所有外露的边线都需要修圆。

1. 单击特征 / Features 工具栏中的圆角 / Fillet。

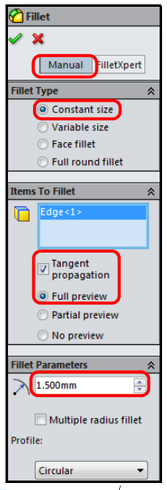
2. 选择图示边线。图中 Select this edge＝选择此边线，箭头指向锚盘顶面最外侧的圆形边线。

ENGLISH ORIGINAL

3 Type **1.5mm** for the fillet radius.

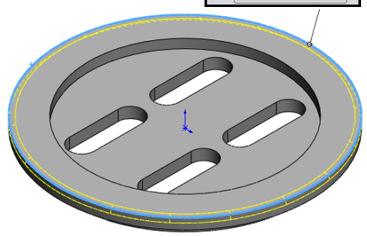
4 The following should be set by default. If not, change them as follows:

- **Manual**
- Fillet Type - **Constant radius**
- Tangent propagation - **Selected**
- Full preview - **Selected**



5 Preview. Once **Full preview** is selected, the outline of the fillet will be shown in yellow.

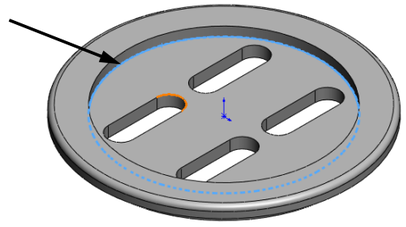
The fillet radius will be shown in a callout, Radius: 1.5mm attached to the edge. Click ✓.



Task 15 — Add a fillet to the inside edge

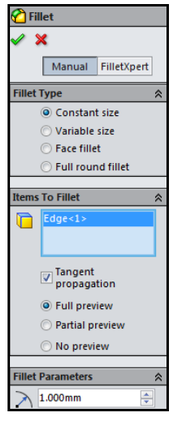
1 Click **Fillet** on the **Features** toolbar.

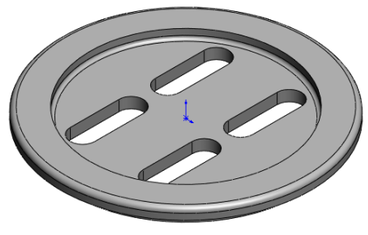
2 Select the edge shown.



3 Type **1mm** for the fillet radius.

4 Click ✓. The inside edge now has a fillet.





中文译文 / CHINESE TRANSLATION

3. 将圆角半径输入为 1.5 mm。

4. 下列选项通常已按默认值设置；若不一致，请改为：手动 / Manual；圆角类型 / Fillet Type — 等半径 / Constant radius；选中切线延伸 / Tangent propagation；选中完整预览 / Full preview。

5. 预览 / Preview：选中完整预览 / Full preview 后，将以黄色显示圆角轮廓。圆角半径以附着于边线的标注显示（Radius: 1.5 mm＝半径：1.5 mm）。单击确定 / OK（绿色对勾）。

任务 15 — 为内侧边线添加圆角 / Add a fillet to the inside edge

1. 单击特征 / Features 工具栏中的圆角 / Fillet。

2. 选择图示边线。图中 Select this edge＝选择此边线，箭头指向凹槽底面与内侧圆柱壁相交的圆形边线。

3. 将圆角半径输入为 1 mm。

4. 单击确定 / OK（绿色对勾）。内侧边线现在已添加圆角。

ENGLISH ORIGINAL

Editing Features

After a feature is created, it can easily be changed. For the extrudes and cuts we made, each end condition or depth could be changed to reflect changes in our design intent, or changes required by later analysis.

For the fillet just created, we could add additional edges to the feature or change the fillet radius.

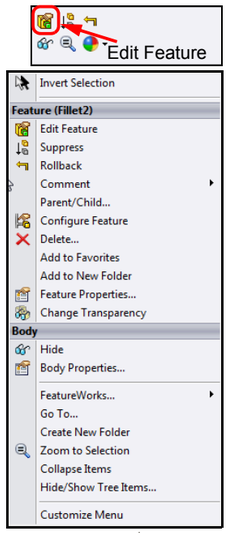
To edit a feature, right-click the feature either in the graphics area or FeatureManager design tree and select **Edit Feature**.

Task 16 — Edit the Fillet

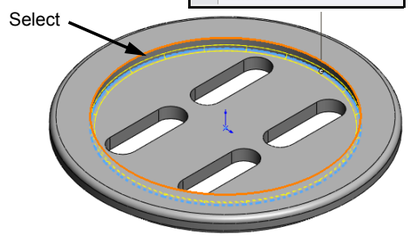
1 In the FeatureManager design tree, right-click the feature `Fillet2` and select **Edit Feature** from the Context toolbar.

SolidWorks can be customized to show the Context toolbar with a text menu below it or just a text menu.

The PropertyManager will now show the same information as when we first applied the fillet.



2 Select the edge shown.



Note: One of these edges will require a round while the other will require a fillet. Both are being done in the same command.

中文译文 / CHINESE TRANSLATION

编辑特征 / Editing Features

特征创建后可以很容易地更改。对于已经创建的拉伸和切除，可以修改其终止条件或深度，以反映设计意图的变化，或满足后续分析提出的更改要求。

对于刚创建的圆角，可以向该特征添加其他边线，也可以更改圆角半径。

要编辑特征，请在图形区域或 FeatureManager 设计树中右键单击该特征，然后选择编辑特征 / Edit Feature。

任务 16 — 编辑圆角 / Edit the Fillet

1. 在 FeatureManager 设计树中右键单击圆角2 / Fillet2 特征，并从上下文工具栏 / Context toolbar 中选择编辑特征 / Edit Feature。

SOLIDWORKS 可以自定义为显示带有下方文字菜单的上下文工具栏，也可以只显示文字菜单。

此时，属性管理器 / PropertyManager 将显示与首次创建该圆角时相同的信息。

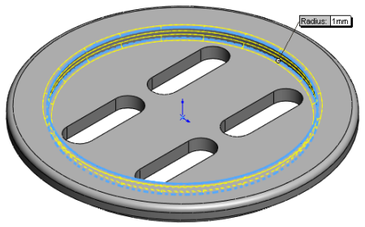
2. 选择图示边线。图中 Select＝选择；箭头指向凹槽内侧圆柱壁顶端的圆形边线，将它加入当前圆角特征的边线选择。

注意 / Note：这两条边线中的一条需要创建外圆角 / Round，另一条需要创建内圆角 / Fillet。两者均在同一命令中完成。

ENGLISH ORIGINAL

3 Once selected, the preview will show that the edge will be rounded as part of the `Fillet2` feature. Click ✔.

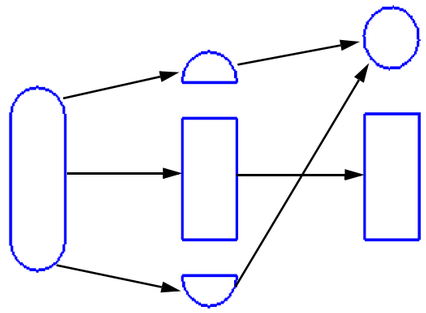
4 **Save** the part.



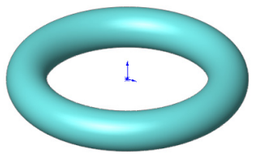
How much does it weight now?

The same principle used earlier of adding and subtracting volumes still applies, however it is now more complicated.

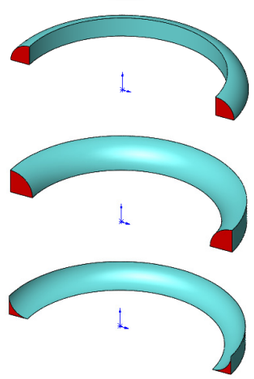
The volume removed by the slots is not too difficult as the area of each slot can be thought of as a rectangle and circle



The fillets are more complicated. The two rounded corners are each part of a torus (donut).



The volumes of the rounds are parts of the torus. The section views at the right show what the two would look like. The volumes are ***not*** one-quarter of the volume of the torus. The equations to determine their volumes are available in both engineering and mathematics handbooks.



The inside corner fillet is even more complicated but still solvable by looking up the equation.

中文译文 / CHINESE TRANSLATION

3. 选中后，预览会显示这条边线将作为圆角2 / Fillet2 特征的一部分被修圆。单击确定 / OK（绿色对勾）。图中半径标注为 Radius: 1 mm（半径：1 mm）。

4. 保存 / Save 零件。

现在的重量是多少？ / How much does it weigh now?

前面使用的体积相加和相减原理仍然适用，但现在计算会更加复杂。

槽所去除的体积并不难计算，因为每个槽的面积可以看作一个矩形加上一个圆。图示将槽的两个半圆端部合并为一个完整的圆。

圆角的计算更复杂。两个修圆的转角各自都是圆环体 / Torus（甜甜圈形状）的一部分。

这些外圆角的体积均为圆环体的部分体积。右侧的剖视图显示了两者的形状；它们的体积并不等于完整圆环体体积的四分之一。用于求这些体积的方程可以在工程手册和数学手册中查到。

内侧转角圆角的计算更加复杂，不过仍然可以通过查找相应方程求解。

ENGLISH ORIGINAL

Using SolidWorks To Get The Weight

Rather than manually solve for the volume of our part and lookup the density of the material, SolidWorks provides tools to solve for the volume and weight of the part.

Add Material

SolidWorks provides a library of materials that can be assigned to parts. Once a material is applied, it will be used by SolidWorks to calculate the weight of the part.

Adding material to a part also changes its visual properties (what it looks like) and graphic properties like the crosshatch used in drawings.

The material assigned to the part can also be used for stress analysis and for photorealistic rendering.

To assign a Material to a part:

- Click  on the **Standard** toolbar.
- Click **Edit, Appearance, Material** in the menu.
- Right-click **Material** in the FeatureManager design tree and select **Edit Material**.

Task 17 — Add material to the part

We are going to manufacture this part from Aluminum 2015.

1 Click **Edit, Appearance, Material** from the menu.
The **Materials Editor** will open.

2 Click the Plus sign next to **Aluminum Alloys** to expand the list.

3 Select **2015 Alloy**.
Select **Metric (MKS)** for **Units**.

中文译文 / CHINESE TRANSLATION

使用 SOLIDWORKS 获取重量 / Using SolidWorks To Get The Weight

不必手工计算零件体积再查找材料密度，SOLIDWORKS 提供了直接求解零件体积和重量的工具。

添加材料 / Add Material

SOLIDWORKS 提供了可以指定给零件的材料库。应用材料后，SOLIDWORKS 将使用该材料计算零件重量。

为零件添加材料还会改变其视觉属性（外观），以及工程图中剖面线等图形属性。

指定给零件的材料也可以用于应力分析和逼真渲染。

为零件指定材料有以下方法：① 单击标准 / Standard 工具栏中图示的材料按钮。② 从菜单选择编辑 / Edit → 外观 / Appearance → 材料 / Material。③ 在 FeatureManager 设计树中右键单击材料 / Material，选择编辑材料 / Edit Material。

任务 17 — 为零件添加材料 / Add material to the part

我们将用 2015 铝合金 / Aluminum 2015 制造此零件。

1. 从菜单选择编辑 / Edit → 外观 / Appearance → 材料 / Material。材料编辑器 / Materials Editor 将打开。

2. 单击铝合金 / Aluminum Alloys 旁的加号，展开列表。

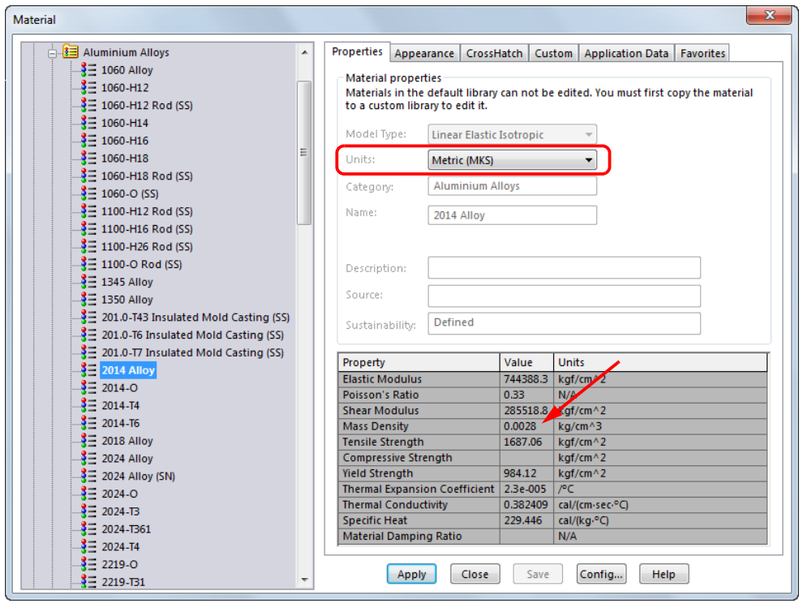
3. 选择 2015 Alloy（2015 合金）。将单位 / Units 选择为公制（MKS）/ Metric (MKS)。

译注（原教材内部差异）：本页正文写的是 2015 Alloy，但下一页的材料库截图及特征树显示的是 2014 Alloy。此处按正文原样翻译；两处不一致应在实际选材时核对。

ENGLISH ORIGINAL

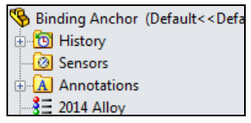
Examine the Physical Properties. Density is listed as .0028 g/mm^3. This is the same value we determined earlier.

Note: Certain graphics cards support RealView advanced graphics visual properties. To find out if yours does, consult the **Help** documentation inside SolidWorks.



4 Click **Apply** and then **Close** to apply the material.

The material is now listed in the FeatureManager design tree.



Mass Properties

Physical properties of a part can easily be calculated using the **Mass Properties** tool. This tool will not only calculate the volume and weight of the part, but many other properties needed during the design and analysis of a part.

To calculate Mass properties:

- Click **Tools, Mass Properties** from the menu
- Or, click on the Tools toolbar

中文译文 / CHINESE TRANSLATION

查看物理属性 / Physical Properties。密度 / Density 列为 0.0028 g/mm^3，与前面计算得到的值相同。

注意 / Note：某些显卡支持 RealView 高级图形视觉效果。要确认你的显卡是否支持，请查阅 SOLIDWORKS 内的帮助 / Help 文档。

材料对话框图示主要标签：Material＝材料；Properties＝属性；Appearance＝外观；CrossHatch＝剖面线；Custom＝自定义；Application Data＝应用程序数据；Favorites＝收藏夹；Material properties＝材料属性；Model Type＝模型类型；Linear Elastic Isotropic＝线性弹性各向同性；Units＝单位；Metric (MKS)＝公制（MKS）；Category＝类别；Name＝名称；Description＝说明；Source＝来源；Sustainability＝可持续性；Defined＝已定义。

图中提示：默认库中的材料不可直接编辑，必须先将材料复制到自定义库后才能编辑。图中所选材料名称为 2014 Alloy（与第 51 页正文的 2015 Alloy 不同）。

图中材料属性标签：Elastic Modulus＝弹性模量；Poisson's Ratio＝泊松比；Shear Modulus＝剪切模量；Mass Density＝质量密度；Tensile Strength＝抗拉强度；Compressive Strength＝抗压强度；Yield Strength＝屈服强度；Thermal Expansion Coefficient＝热膨胀系数；Thermal Conductivity＝热导率；Specific Heat＝比热；Material Damping Ratio＝材料阻尼比；Value＝数值；Units＝单位。

4. 单击应用 / Apply，然后单击关闭 / Close，以应用材料。该材料现在会列在 FeatureManager 设计树中。

质量属性 / Mass Properties

使用质量属性 / Mass Properties 工具，可以很容易地计算零件的物理属性。该工具不仅计算零件体积和重量，还会计算零件设计与分析所需的许多其他属性。

计算质量属性：从菜单选择工具 / Tools → 质量属性 / Mass Properties；或单击工具 / Tools 工具栏中的图示质量属性按钮。

ENGLISH ORIGINAL

Task 18 — Determine The Weight

1 Click **Tools, Mass Properties** from the menu.

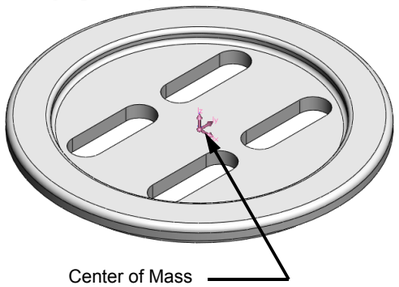
The Mass Properties box will appear.

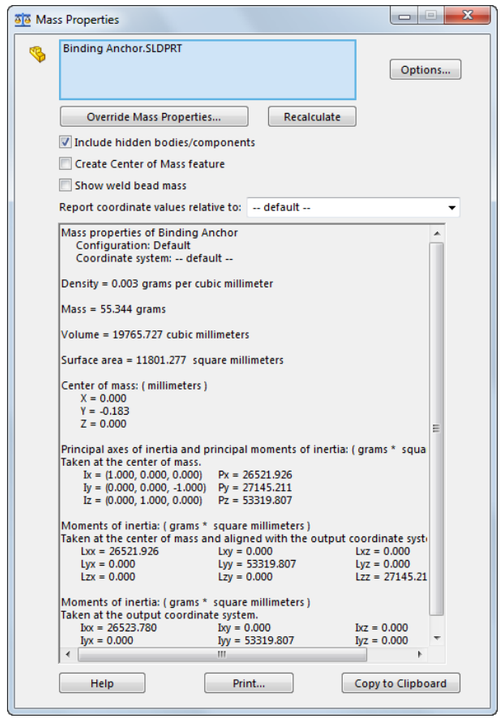
The Volume is calculated to be **19,765.727mm^3** and the Mass is **55.344 grams**.

When we calculated the weight earlier it was 62.68 grams. This was before we removed material with the four slots and two rounds.

2 The Center of Mass is the balance point of the part. If we could suspend the part at this point, it would not want to tip over.

The center of mass is displayed numerically in the box and graphically by a pink triad.





中文译文 / CHINESE TRANSLATION

任务 18 — 确定重量 / Determine The Weight

1. 从菜单选择工具 / Tools → 质量属性 / Mass Properties。质量属性 / Mass Properties 对话框将显示。

计算得到的体积 / Volume 为 19,765.727 mm³，质量 / Mass 为 55.344 g。

前面计算得到的重量为 62.68 g。当时尚未通过四个槽和两个外圆角去除材料。

2. 质心 / Center of Mass 是零件的平衡点。如果能够在此点悬挂零件，零件就不会倾斜翻转。

质心既以数值形式显示在对话框中，也以粉红色三轴标记显示在图形区域中。图中的 Center of Mass＝质心。

截图主要标签：Options＝选项；Override Mass Properties＝覆盖质量属性；Recalculate＝重新计算；Include hidden bodies/components＝包括隐藏的实体/零部件；Create Center of Mass feature＝创建质心特征；Show weld bead mass＝显示焊缝质量；Report coordinate values relative to＝报告坐标值相对于；Density＝密度；Surface area＝表面积；Principal axes of inertia and principal moments of inertia＝惯性主轴和主惯性矩；Moments of inertia＝惯性矩；Help＝帮助；Close＝关闭；Print＝打印；Copy to Clipboard＝复制到剪贴板。

截图中其他显示值：表面积 11,801.277 mm²；质心坐标 X＝0.000 mm、Y＝−0.183 mm、Z＝0.000 mm。密度在此截图中按显示精度取整为 0.003 g/mm³。

ENGLISH ORIGINAL

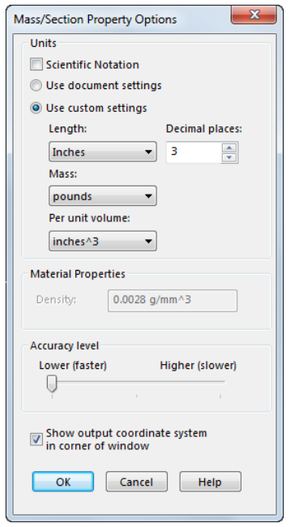
Change Units.

The units for the `Binding Anchor` part are in millimeters, grams, seconds, so the mass properties displayed in millimeters and grams. If we need the mass properties in different units, such as inch, pound, second, the conversion is simple.

3 Click the **Options** button.

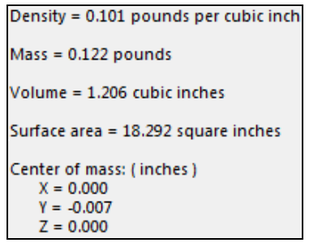
4 Select **Use custom settings**. Select:

- Length - **Inches**
- Mass - **Pounds**
- Per unit volume - **inches^3**



5 Click **OK**.

Note: The units have only been changed in this output. The part still uses millimeters as the unit of length



6 Click **Close** to close the Mass Properties.

7 **Save** the part.

中文译文 / CHINESE TRANSLATION

更改单位 / Change Units

固定器锚盘 / Binding Anchor 零件使用毫米、克、秒单位制，因此质量属性以毫米和克显示。如果需要使用其他单位（例如英寸、磅、秒）显示质量属性，转换非常简单。

3. 单击选项 / Options 按钮。

4. 选择使用自定义设置 / Use custom settings，并选择：长度 / Length — 英寸 / Inches；质量 / Mass — 磅 / Pounds；单位体积 / Per unit volume — 立方英寸 / inches^3。

5. 单击确定 / OK。

注意 / Note：这里只更改该输出的单位；零件本身的长度单位仍然是毫米。

6. 单击关闭 / Close，关闭质量属性 / Mass Properties 对话框。

7. 保存 / Save 零件。

截图中其他选项标签：Mass/Section Property Options＝质量/截面属性选项；Scientific Notation＝科学记数法；Use document settings＝使用文档设置；Decimal places＝小数位数；Material Properties＝材料属性；Accuracy level＝精度等级；Lower (faster)＝较低（更快）；Higher (slower)＝较高（更慢）；Show output coordinate system in corner of window＝在窗口角落显示输出坐标系。

转换后的图示结果：密度 / Density＝0.101 磅/立方英寸；质量 / Mass＝0.122 磅；体积 / Volume＝1.206 立方英寸；表面积 / Surface area＝18.292 平方英寸；质心 / Center of mass（英寸）X＝0.000、Y＝−0.007、Z＝0.000。